

28	An electron in an atom transitions from -25.0 eV to -80.0 eV . A photon is emitted in the process. What is the wavelength of the emitted photon?							
	A	15.5 nm	B	22.6 nm	C	49.7 nm	D	300 nm
	Answer: B Explanation/Working: Energy difference							

[Turn over

$$= -80.0 - (-25.0) \text{ eV} = -55.0 \text{ eV}$$

$$= -55.0 \times 1.60 \times 10^{-19} \text{ J}$$

$$= 8.81 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{8.81 \times 10^{-18}} = 2.26 \times 10^{-8} \text{ m}$$

$$\lambda = 22.6 \text{ nm}$$